

VSDBabySoC Static Timing Analysis

STA Report Across SKY130 HD Timing Corners

Assessment Date	19/09/2026
Design	VSDBabySoC
Top Module	vsdbabysoc
Timing Analysis Tool	OpenSTA
Technology	SKY130 HD
Timing Corners	18

Static Timing Analysis Report

1. Introduction

Static Timing Analysis (STA) is a method used to verify the timing performance of a digital design without applying functional simulation vectors. STA checks timing paths between sequential and combinational elements and determines whether setup and hold timing requirements are met. In this work, OpenSTA was used to analyze the synthesized VSDBabySoC netlist using SKY130 HD standard-cell timing libraries. The analysis was performed for multiple process, voltage and temperature corners.

2. Design and STA Files

Item	Path
Synthesized Netlist	/home/vsduser/OpenSTA/BabySoC_Simulation/src/vsdbabysoc.synth.v
SDC	/home/vsduser/Desktop/VSDBabySoC/src/sdc/vsdbabysoc_synthesis.sdc
PLL Library	/home/vsduser/Desktop/VSDBabySoC/src/lib/avsdpll.lib
DAC Library	/home/vsduser/Desktop/VSDBabySoC/src/lib/avsdDAC.lib
Timing Libraries	/home/vsduser/Desktop/skywater-pdk-libs-sky130_fd_sc_hd/timing/

3. Timing Corner Meaning

Library	Process	Temperature	Voltage
ff_100C_1v65	Fast-Fast	100°C	1.65 V
ff_100C_1v95	Fast-Fast	100°C	1.95 V
ff_n40C_1v56	Fast-Fast	-40°C	1.56 V
ff_n40C_1v65	Fast-Fast	-40°C	1.65 V
ff_n40C_1v76	Fast-Fast	-40°C	1.76 V
ff_n40C_1v95	Fast-Fast	-40°C	1.95 V
ff_n40C_1v95_ccsnoise	Fast-Fast + CCS noise	-40°C	1.95 V
ss_100C_1v40	Slow-Slow	100°C	1.40 V
ss_100C_1v60	Slow-Slow	100°C	1.60 V
ss_n40C_1v28	Slow-Slow	-40°C	1.28 V
ss_n40C_1v35	Slow-Slow	-40°C	1.35 V
ss_n40C_1v40	Slow-Slow	-40°C	1.40 V
ss_n40C_1v44	Slow-Slow	-40°C	1.44 V
ss_n40C_1v60	Slow-Slow	-40°C	1.60 V
ss_n40C_1v60_ccsnoise	Slow-Slow + CCS noise	-40°C	1.60 V
ss_n40C_1v76	Slow-Slow	-40°C	1.76 V
tt_025C_1v80	Typical-Typical	25°C	1.80 V
tt_100C_1v80	Typical-Typical	100°C	1.80 V

4. STA Results

Corner	Min Slack (ns)	Max Slack (ns)	Hold	Setup
ff_100C_1v65	0.25	4.05	MET	MET
ff_100C_1v95	0.20	5.57	MET	MET
ff_n40C_1v56	0.29	2.28	MET	MET
ff_n40C_1v65	0.26	3.41	MET	MET
ff_n40C_1v76	0.22	4.43	MET	MET
ff_n40C_1v95	0.19	5.59	MET	MET
ff_n40C_1v95_ccsnoise	0.19	5.59	MET	MET
ss_100C_1v40	0.91	-17.12	MET	VIOLATED
ss_100C_1v60	0.64	-7.87	MET	VIOLATED
ss_n40C_1v28	1.83	-62.56	MET	VIOLATED
ss_n40C_1v35	1.35	-39.70	MET	VIOLATED
ss_n40C_1v40	1.12	-29.69	MET	VIOLATED
ss_n40C_1v44	0.99	-23.91	MET	VIOLATED
ss_n40C_1v60	0.66	-10.61	MET	VIOLATED
ss_n40C_1v60_ccsnoise	0.66	-10.61	MET	VIOLATED
ss_n40C_1v76	0.50	-4.38	MET	VIOLATED
tt_025C_1v80	0.31	1.94	MET	MET
tt_100C_1v80	0.31	2.09	MET	MET

5. Analysis Methodology

The STA flow loads the required Liberty timing libraries, reads and links the synthesized VSDBabySoC netlist, applies the SDC constraints, and executes OpenSTA checks for each timing corner. Corner-specific reports are stored and processed to extract WNS, TNS and minimum slack values.

6. Key Results

The supplied results show positive hold slack across all listed corners. Setup timing is met for the FF and TT corners shown, while several SS corners exhibit negative maximum slack. The most severe displayed setup result is approximately –62.56 ns at SS n40C 1.28 V.

Figure 1. OpenSTA Timing Analysis

```
vsduser@vdsquadron:~/OpenSTA$ docker run -it -v $HOME:/data opensta
OpenSTA 2.4.0 ac3479bc24 Copyright (c) 2021, Parallax Software, Inc.
License GPLv3: GNU GPL version 3 <http://gnu.org/licenses/gpl.html>

This is free software, and you are free to change and redistribute it
under certain conditions; type 'show_copying' for details.
This program comes with ABSOLUTELY NO WARRANTY; for details type 'show_warranty'.
% read_liberty /OpenSTA/examples/nangate45_slow.lib
1
% read_verilog /OpenSTA/examples/example1.v
1
% link_design top
1
% create_clock -name clk -period 10 {clk1 clk2 clk3}
% set_input_delay -clock clk 0 {in1 in2}
% report_checks
Startpoint: r2 (rising edge-triggered flip-flop clocked by clk)
Endpoint: r3 (rising edge-triggered flip-flop clocked by clk)
Path Group: clk
Path Type: max

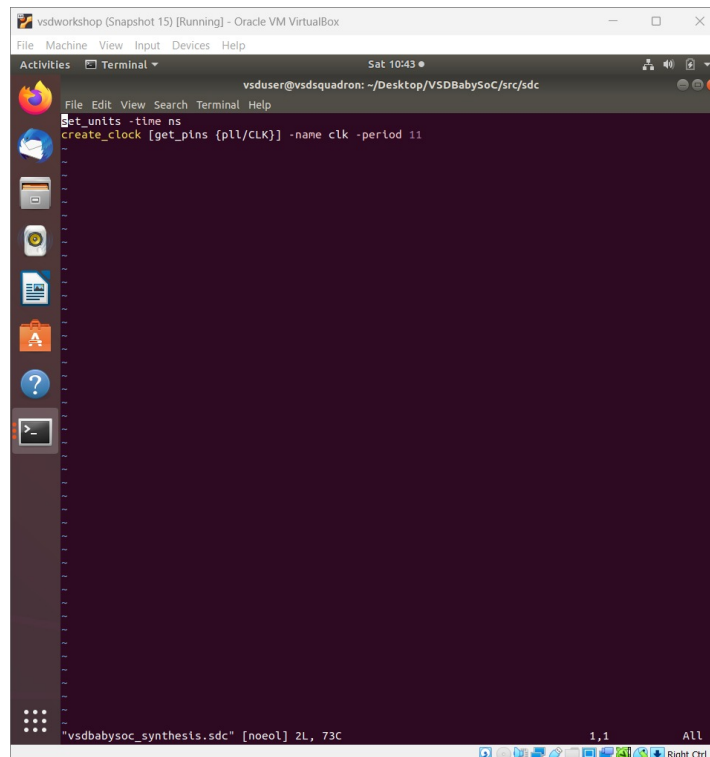
Delay    Time    Description
-----
0.00     0.00    clock clk (rise edge)
0.00     0.00    clock network delay (ideal)
0.00     0.00    ^ r2/CK (DFF_X1)
0.23     0.23    v r2/Q (DFF_X1)
0.08     0.31    v u1/Z (BUF_X1)
0.10     0.41    v u2/ZN (AND2_X1)
0.00     0.41    v r3/D (DFF_X1)
          0.41    data arrival time

10.00    10.00    clock clk (rise edge)
0.00     10.00    clock network delay (ideal)
0.00     10.00    clock reconvergence pessimism
10.00    10.00    ^ r3/CK (DFF_X1)
-0.16    9.84    library setup time
          9.84    data required time
-----
          9.84    data required time
-0.41    -0.41    data arrival time
-----
          9.43    slack (MET)

%
```

OpenSTA timing path analysis showing arrival time, required time and slack.

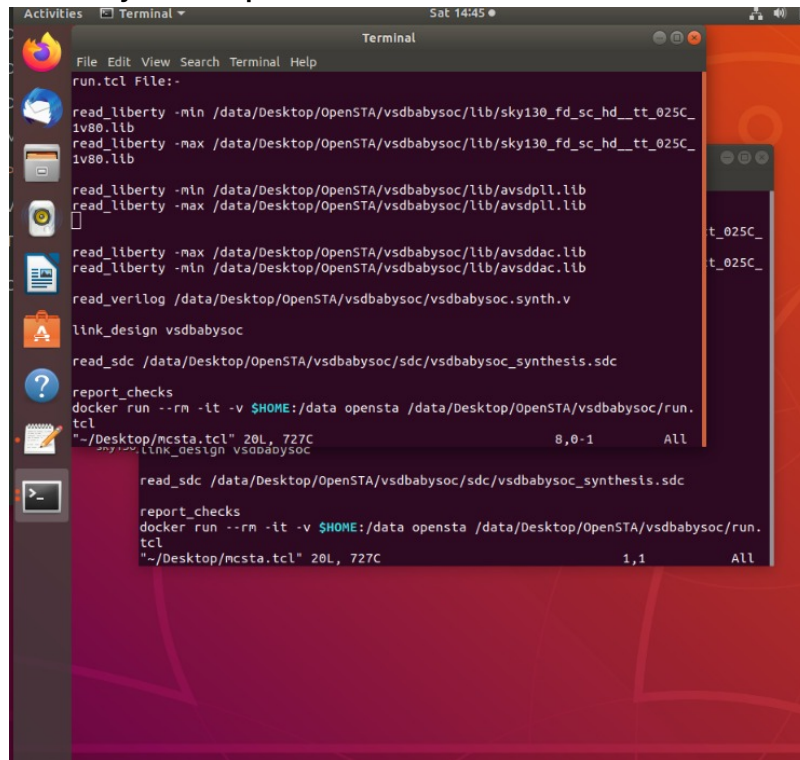
Figure 2. SDC Timing Constraint File

A screenshot of a terminal window titled "vsdworkshop (Snapshot 15) [Running] - Oracle VM VirtualBox". The terminal shows the user "vsduser@vdsquadron" at the prompt. The current directory is "~/Desktop/VSDBabySoC/src/sdc". The user has entered two commands: "set_units -time ns" and "create_clock [get_pins {pll1/CLK}] -name clk -period 11". The terminal output shows the commands being executed. The window also displays a sidebar with application icons and a status bar at the bottom indicating the file "vsdbabysoc_synthesis.sdc" is open at line 1, column 1.

```
vsduser@vdsquadron: ~/Desktop/VSDBabySoC/src/sdc
set_units -time ns
create_clock [get_pins {pll1/CLK}] -name clk -period 11
```

SDC clock and timing constraints used for the VSDBabySoC STA flow.

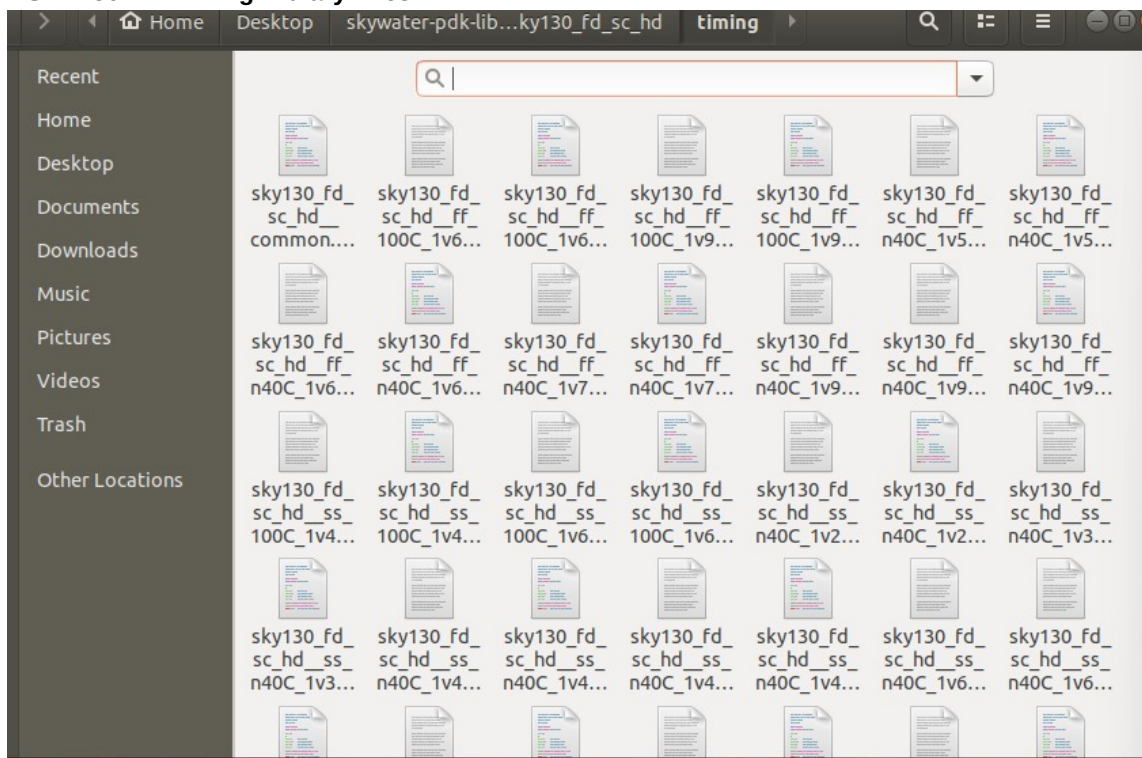
Figure 3. OpenSTA Multi-Library Run Script



```
run.tcl File:-
read_liberty -min /data/Desktop/OpenSTA/vsdbabysoc/lib/sky130_fd_sc_hd_tt_025C_1v80.lib
read_liberty -max /data/Desktop/OpenSTA/vsdbabysoc/lib/sky130_fd_sc_hd_tt_025C_1v80.lib
read_liberty -min /data/Desktop/OpenSTA/vsdbabysoc/lib/avsdpll.lib
read_liberty -max /data/Desktop/OpenSTA/vsdbabysoc/lib/avsdpll.lib
read_liberty -max /data/Desktop/OpenSTA/vsdbabysoc/lib/avsdac.lib
read_liberty -min /data/Desktop/OpenSTA/vsdbabysoc/lib/avsdac.lib
read_verilog /data/Desktop/OpenSTA/vsdbabysoc/vsdbabysoc.synth.v
link_design vsdbabysoc
read_sdc /data/Desktop/OpenSTA/vsdbabysoc/sdc/vsdbabysoc_synthesis.sdc
report_checks
docker run --rm -it -v $HOME:/data opensta /data/Desktop/OpenSTA/vsdbabysoc/run.tcl
"/Desktop/mcsta.tcl" 20L, 727C 8,0-1 All
link_design vsdbabysoc
read_sdc /data/Desktop/OpenSTA/vsdbabysoc/sdc/vsdbabysoc_synthesis.sdc
report_checks
docker run --rm -it -v $HOME:/data opensta /data/Desktop/OpenSTA/vsdbabysoc/run.tcl
"/Desktop/mcsta.tcl" 20L, 727C 1,1 All
```

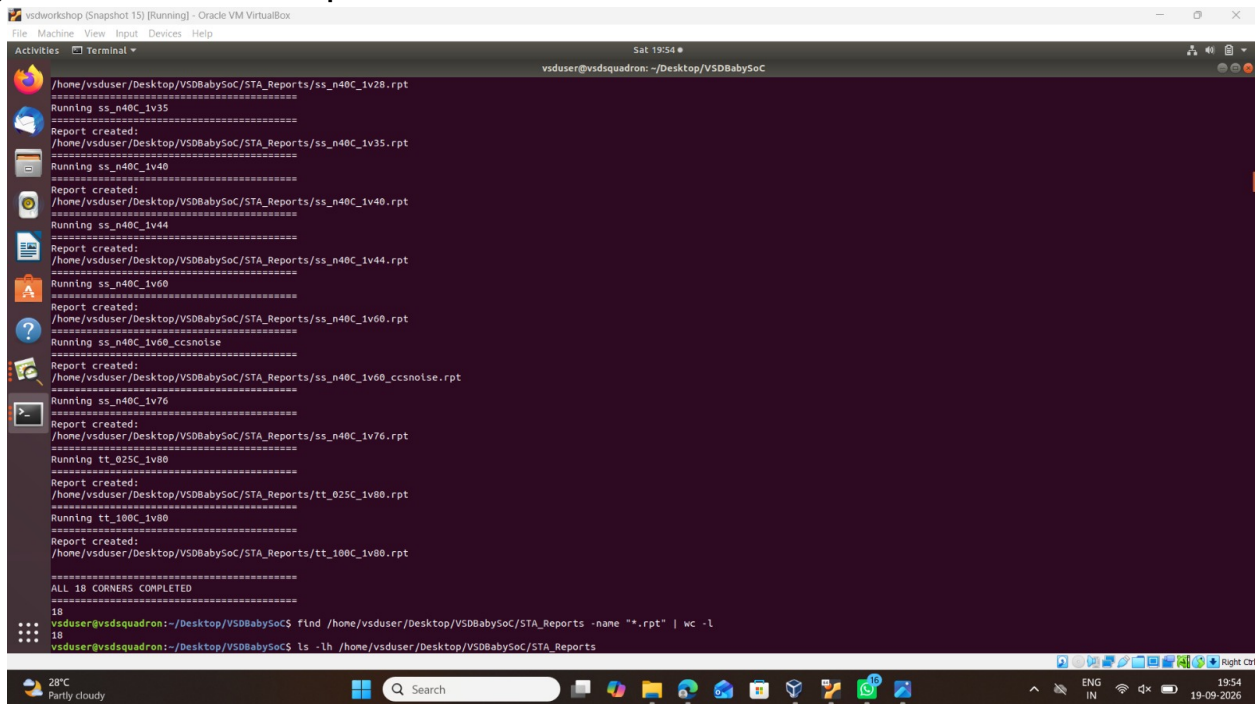
Libraries, synthesized netlist and SDC constraints loaded before timing checks.

Figure 4. SKY130 HD Timing Library Files



SKY130 HD characterized timing libraries used for multi-corner analysis.

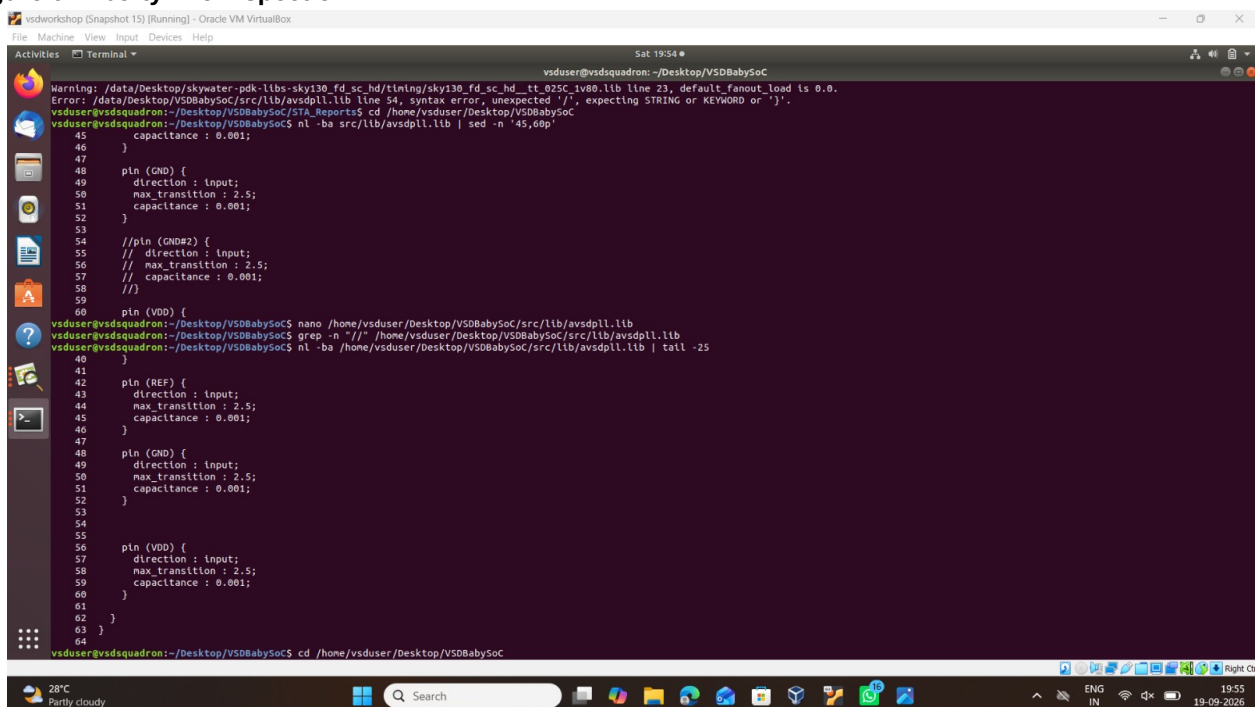
Figure 5. Multi-Corner STA Report Generation



```
vduser@vdsquadron: ~/Desktop/VSDBabySoC
=====
Running ss_n40C_1v28.rpt
Report created:
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ss_n40C_1v28.rpt
=====
Running ss_n40C_1v35.rpt
Report created:
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ss_n40C_1v35.rpt
=====
Running ss_n40C_1v40.rpt
Report created:
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ss_n40C_1v40.rpt
=====
Running ss_n40C_1v44.rpt
Report created:
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ss_n40C_1v44.rpt
=====
Running ss_n40C_1v60.rpt
Report created:
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ss_n40C_1v60.rpt
=====
Running ss_n40C_1v60_ccsnolse.rpt
Report created:
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ss_n40C_1v60_ccsnolse.rpt
=====
Running ss_n40C_1v76.rpt
Report created:
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ss_n40C_1v76.rpt
=====
Running tt_025C_1v80.rpt
Report created:
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/tt_025C_1v80.rpt
=====
Running tt_100C_1v80.rpt
Report created:
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/tt_100C_1v80.rpt
=====
ALL 18 CORNERS COMPLETED
=====
vduser@vdsquadron:~/Desktop/VSDBabySoC$ find /home/vsduser/Desktop/VSDBabySoC/STA_Reports -name "*.rpt" | wc -l
18
vduser@vdsquadron:~/Desktop/VSDBabySoC$ ls -lh /home/vsduser/Desktop/VSDBabySoC/STA_Reports
```

Automated generation of individual STA reports for the configured corners.

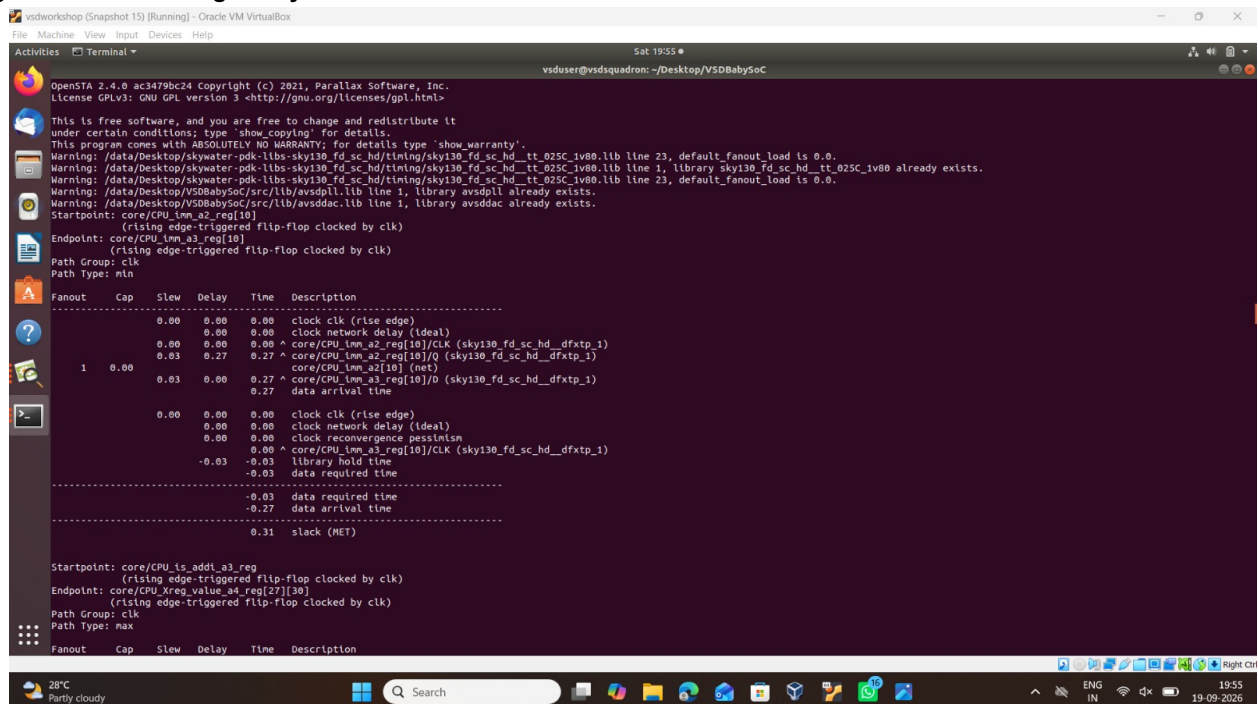
Figure 6. Liberty File Inspection



```
vduser@vdsquadron: ~/Desktop/VSDBabySoC
Warning: /data/Desktop/skywater-pdk-libs-sky130-fd-sc-hd/timing/sky130-fd-sc-hd-tt_025C_1v80.lib line 23, default fanout_load is 0.0.
Error: /data/Desktop/VSDBabySoC/src/lib/avsdpll.lib line 54, syntax error, unexpected '/', expecting STRING or KEYWORD or '}'.
vduser@vdsquadron:~/Desktop/VSDBabySoC/STA_Reports$ cd /home/vsduser/Desktop/VSDBabySoC
vduser@vdsquadron:~/Desktop/VSDBabySoC$ nl -ba src/lib/avsdpll.lib | sed -n '45,60p'
45     capacitance : 0.001;
46 }
47
48 pin (GND) {
49     direction : input;
50     max_transition : 2.5;
51     capacitance : 0.001;
52 }
53
54 //pin (GND#2) {
55 //     direction : input;
56 //     max_transition : 2.5;
57 //     capacitance : 0.001;
58 // }
59
60 pin (VDD) {
vduser@vdsquadron:~/Desktop/VSDBabySoC$ nano /home/vsduser/Desktop/VSDBabySoC/src/lib/avsdpll.lib
vduser@vdsquadron:~/Desktop/VSDBabySoC$ grep -n "/" /home/vsduser/Desktop/VSDBabySoC/src/lib/avsdpll.lib
vduser@vdsquadron:~/Desktop/VSDBabySoC$ nl -ba /home/vsduser/Desktop/VSDBabySoC/src/lib/avsdpll.lib | tail -25
40 }
41
42 pin (REF) {
43     direction : input;
44     max_transition : 2.5;
45     capacitance : 0.001;
46 }
47
48 pin (GND) {
49     direction : input;
50     max_transition : 2.5;
51     capacitance : 0.001;
52 }
53
54
55
56 pin (VDD) {
57     direction : input;
58     max_transition : 2.5;
59     capacitance : 0.001;
60 }
61
62 }
63 }
64
vduser@vdsquadron:~/Desktop/VSDBabySoC$ cd /home/vsduser/Desktop/VSDBabySoC
```

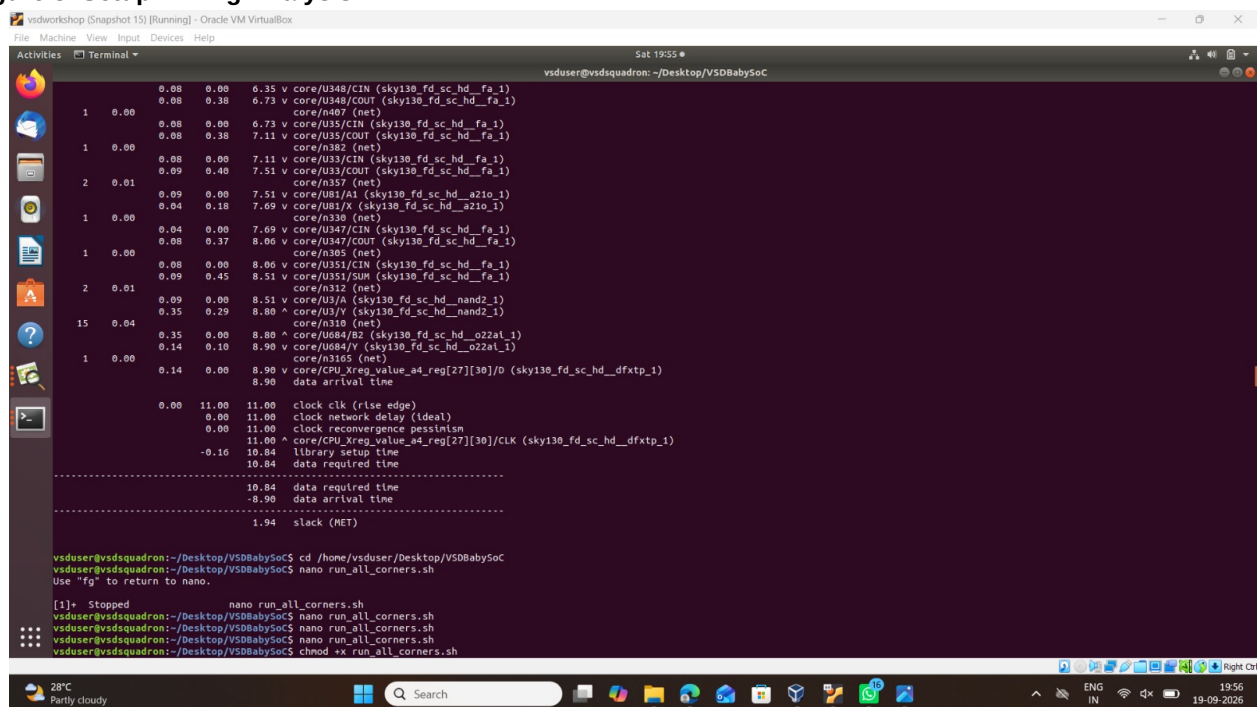
Inspection of custom Liberty library pin and timing definitions.

Figure 7. Hold Timing Analysis



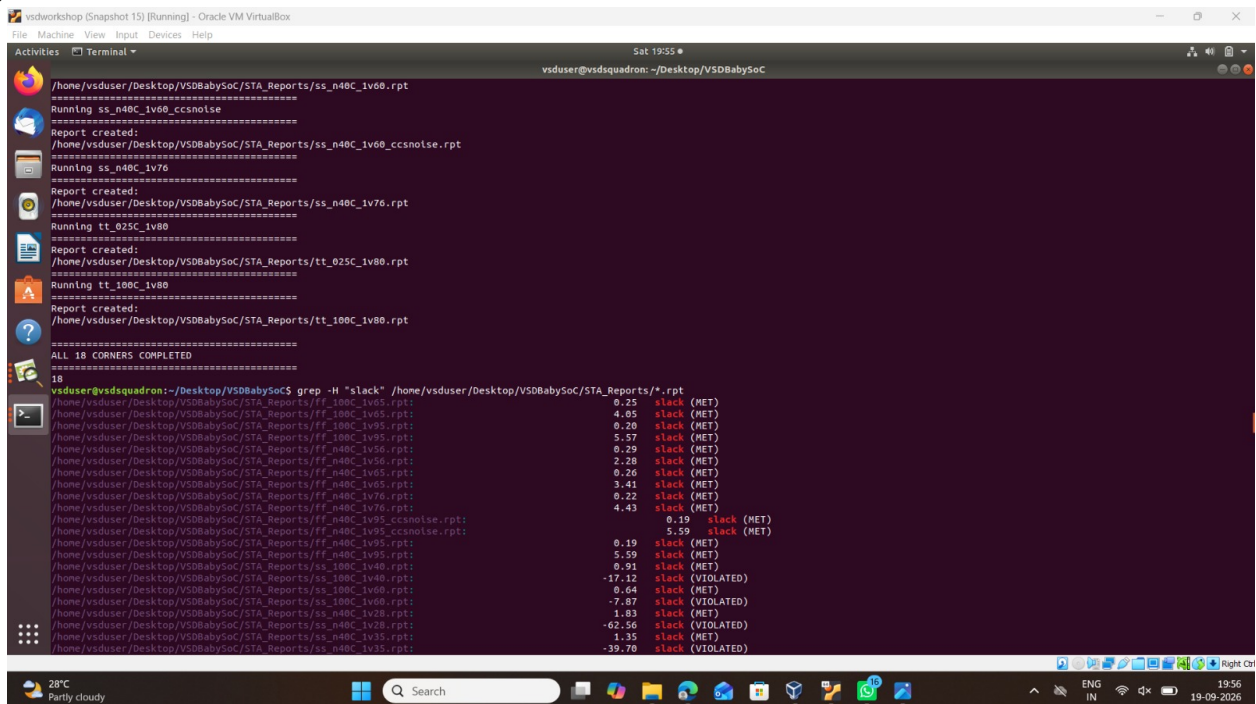
Minimum-delay path with positive hold slack and a MET timing result.

Figure 8. Setup Timing Analysis



Maximum-delay path used for setup timing and slack calculation.

Figure 9. Slack Extraction Across 18 Corners



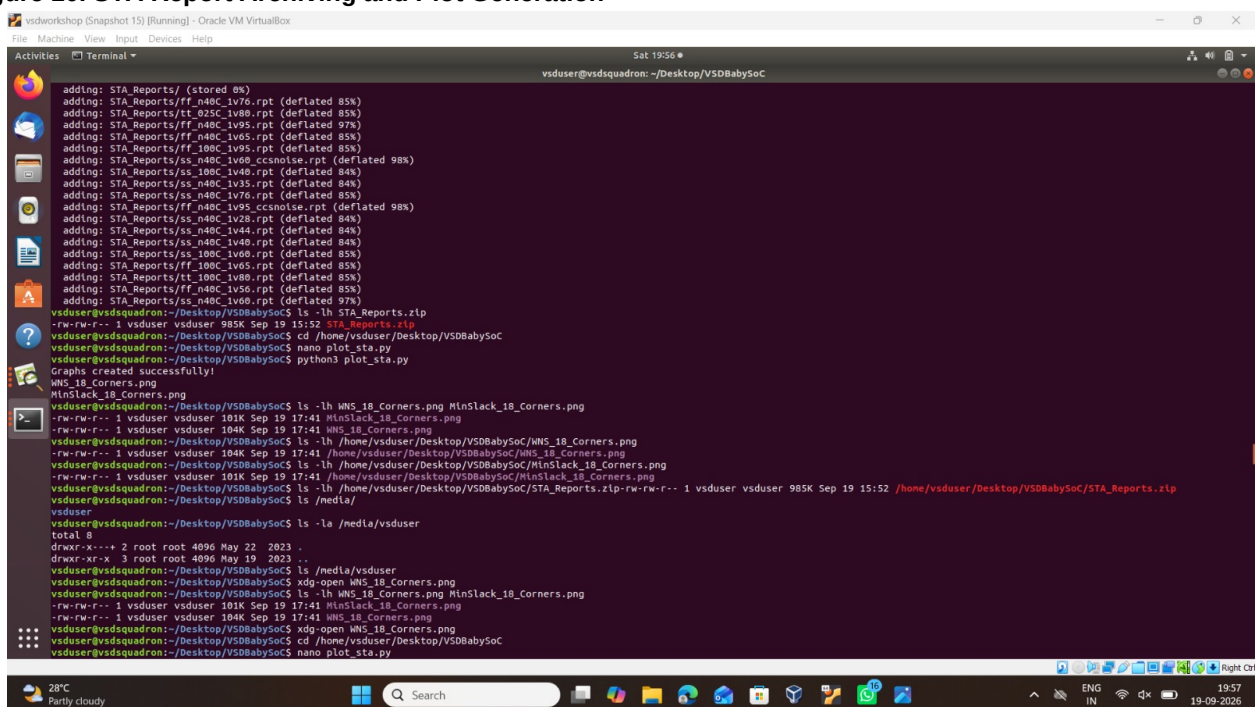
```

/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ss_n40C_iv00.rpt
Running ss_n40C_iv00_ccsnolse
Report created:
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ss_n40C_iv00_ccsnolse.rpt
Running ss_n40C_iv76
Report created:
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ss_n40C_iv76.rpt
Running tt_025C_iv80
Report created:
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/tt_025C_iv80.rpt
Running tt_100C_iv80
Report created:
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/tt_100C_iv80.rpt
=====
ALL 18 CORNERS COMPLETED
=====
18
vsduser@vdsquadron:~/Desktop/VSDBabySoC$ grep -H "slack" /home/vsduser/Desktop/VSDBabySoC/STA_Reports/*.rpt
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ff_n40C_iv05.rpt:0.25 slack (MET)
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ff_n40C_iv05.rpt:4.05 slack (MET)
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ff_n40C_iv95.rpt:0.20 slack (MET)
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ff_n40C_iv95.rpt:5.57 slack (MET)
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ff_n40C_iv95.rpt:0.29 slack (MET)
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ff_n40C_iv95.rpt:2.28 slack (MET)
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ff_n40C_iv05.rpt:0.26 slack (MET)
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ff_n40C_iv05.rpt:3.41 slack (MET)
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ff_n40C_iv76.rpt:0.22 slack (MET)
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ff_n40C_iv76.rpt:4.43 slack (MET)
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ff_n40C_iv95_ccsnolse.rpt:0.19 slack (MET)
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ff_n40C_iv95_ccsnolse.rpt:5.59 slack (MET)
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ff_n40C_iv95.rpt:0.19 slack (MET)
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ff_n40C_iv95.rpt:5.59 slack (MET)
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ss_100C_iv40.rpt:0.91 slack (MET)
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ss_100C_iv40.rpt:-17.12 slack (VIOLATED)
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ss_100C_iv00.rpt:0.64 slack (MET)
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ss_100C_iv00.rpt:-7.87 slack (VIOLATED)
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ss_n40C_iv28.rpt:1.83 slack (MET)
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ss_n40C_iv08.rpt:-62.56 slack (VIOLATED)
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ss_n40C_iv35.rpt:1.35 slack (MET)
/home/vsduser/Desktop/VSDBabySoC/STA_Reports/ss_n40C_iv35.rpt:-39.70 slack (VIOLATED)

```

Corner-wise extraction of slack values from the generated reports.

Figure 10. STA Report Archiving and Plot Generation



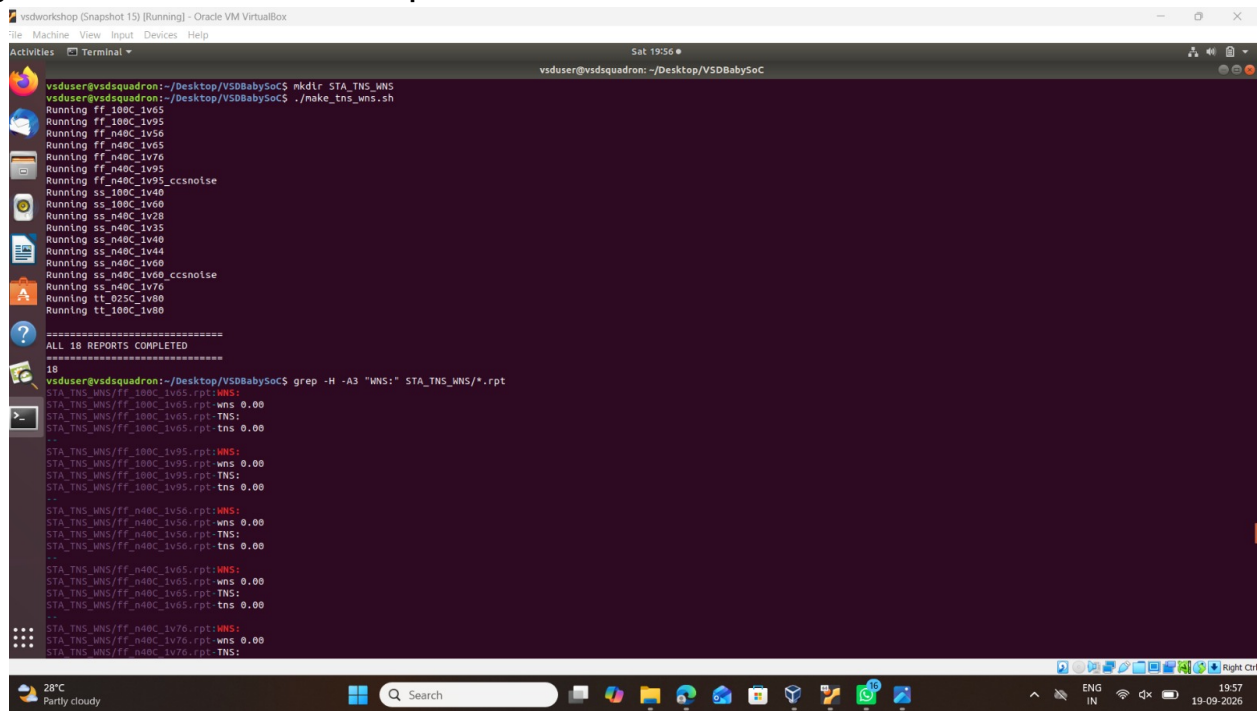
```

adding: STA_Reports/ (stored 0%)
adding: STA_Reports/ff_n40C_iv76.rpt (deflated 85%)
adding: STA_Reports/tt_025C_iv80.rpt (deflated 85%)
adding: STA_Reports/ff_n40C_iv95.rpt (deflated 97%)
adding: STA_Reports/ff_n40C_iv05.rpt (deflated 85%)
adding: STA_Reports/ff_100C_iv95.rpt (deflated 85%)
adding: STA_Reports/ss_n40C_iv00_ccsnolse.rpt (deflated 98%)
adding: STA_Reports/ss_100C_iv40.rpt (deflated 84%)
adding: STA_Reports/ss_n40C_iv35.rpt (deflated 84%)
adding: STA_Reports/ss_n40C_iv76.rpt (deflated 85%)
adding: STA_Reports/ff_n40C_iv95_ccsnolse.rpt (deflated 98%)
adding: STA_Reports/ss_n40C_iv28.rpt (deflated 84%)
adding: STA_Reports/ss_n40C_iv44.rpt (deflated 84%)
adding: STA_Reports/ss_n40C_iv40.rpt (deflated 84%)
adding: STA_Reports/ss_100C_iv00.rpt (deflated 85%)
adding: STA_Reports/ff_100C_iv05.rpt (deflated 85%)
adding: STA_Reports/tt_100C_iv80.rpt (deflated 85%)
adding: STA_Reports/ff_n40C_iv56.rpt (deflated 85%)
adding: STA_Reports/ss_n40C_iv00.rpt (deflated 97%)
vsduser@vdsquadron:~/Desktop/VSDBabySoC$ ls -lh STA_Reports.zip
-rw-rw-r-- 1 vsduser vsduser 985K Sep 19 15:52 STA_Reports.zip
vsduser@vdsquadron:~/Desktop/VSDBabySoC$ cd /home/vsduser/Desktop/VSDBabySoC
vsduser@vdsquadron:~/Desktop/VSDBabySoC$ nano plot_sta.py
vsduser@vdsquadron:~/Desktop/VSDBabySoC$ python3 plot_sta.py
Graphs created successfully!
MnSlack_18_Corners.png
MnSlack_18_Corners.png
vsduser@vdsquadron:~/Desktop/VSDBabySoC$ ls -lh MnSlack_18_Corners.png
-rw-rw-r-- 1 vsduser vsduser 101K Sep 19 17:41 MnSlack_18_Corners.png
-rw-rw-r-- 1 vsduser vsduser 104K Sep 19 17:41 MnSlack_18_Corners.png
vsduser@vdsquadron:~/Desktop/VSDBabySoC$ ls -lh /home/vsduser/Desktop/VSDBabySoC/MnSlack_18_Corners.png
-rw-rw-r-- 1 vsduser vsduser 104K Sep 19 17:41 /home/vsduser/Desktop/VSDBabySoC/MnSlack_18_Corners.png
vsduser@vdsquadron:~/Desktop/VSDBabySoC$ ls -lh /home/vsduser/Desktop/VSDBabySoC/MnSlack_18_Corners.png
-rw-rw-r-- 1 vsduser vsduser 101K Sep 19 17:41 /home/vsduser/Desktop/VSDBabySoC/MnSlack_18_Corners.png
vsduser@vdsquadron:~/Desktop/VSDBabySoC$ ls -lh /home/vsduser/Desktop/VSDBabySoC/STA_Reports.zip
-rw-rw-r-- 1 vsduser vsduser 985K Sep 19 15:52 /home/vsduser/Desktop/VSDBabySoC/STA_Reports.zip
vsduser@vdsquadron:~/Desktop/VSDBabySoC$ cd /media/
vsduser@vdsquadron:~/Desktop/VSDBabySoC$ ls -la /media/vsduser
total 8
drwxr-xr-x 2 root root 4096 May 22 2023 .
drwxr-xr-x 3 root root 4096 May 19 2023 ..
vsduser@vdsquadron:~/Desktop/VSDBabySoC$ ls /media/vsduser
vsduser@vdsquadron:~/Desktop/VSDBabySoC$ open MnSlack_18_Corners.png
vsduser@vdsquadron:~/Desktop/VSDBabySoC$ ls -lh MnSlack_18_Corners.png
-rw-rw-r-- 1 vsduser vsduser 101K Sep 19 17:41 MnSlack_18_Corners.png
-rw-rw-r-- 1 vsduser vsduser 104K Sep 19 17:41 MnSlack_18_Corners.png
vsduser@vdsquadron:~/Desktop/VSDBabySoC$ xdg-open MnSlack_18_Corners.png
vsduser@vdsquadron:~/Desktop/VSDBabySoC$ cd /home/vsduser/Desktop/VSDBabySoC
vsduser@vdsquadron:~/Desktop/VSDBabySoC$ nano plot_sta.py

```

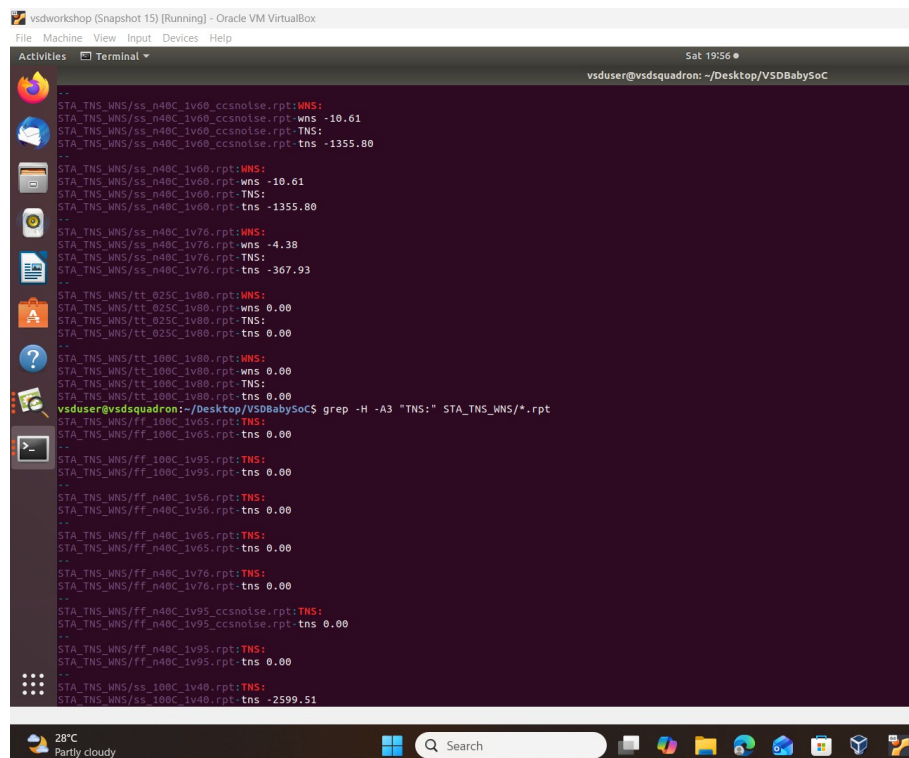
STA report archive and successful generation of timing plot files.

Figure 11. Multi-Corner TNS/WNS Report Generation



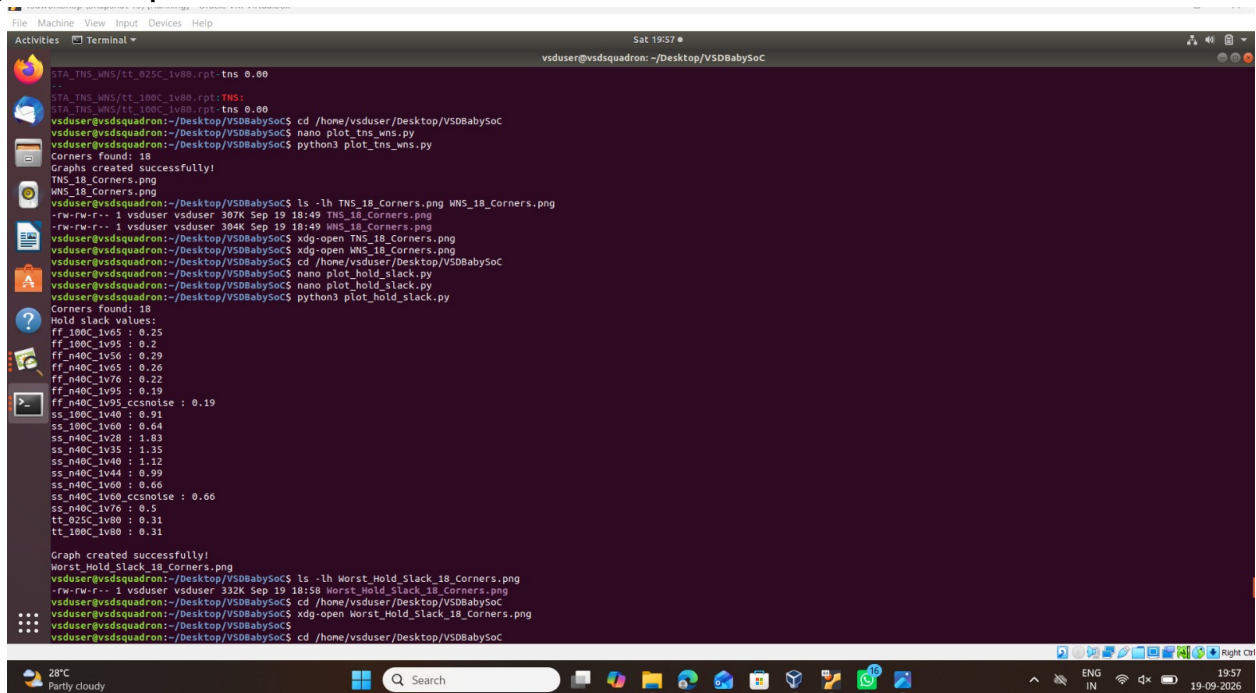
Automated WNS/TNS extraction across all 18 timing corners.

Figure 12. TNS and WNS Value Extraction



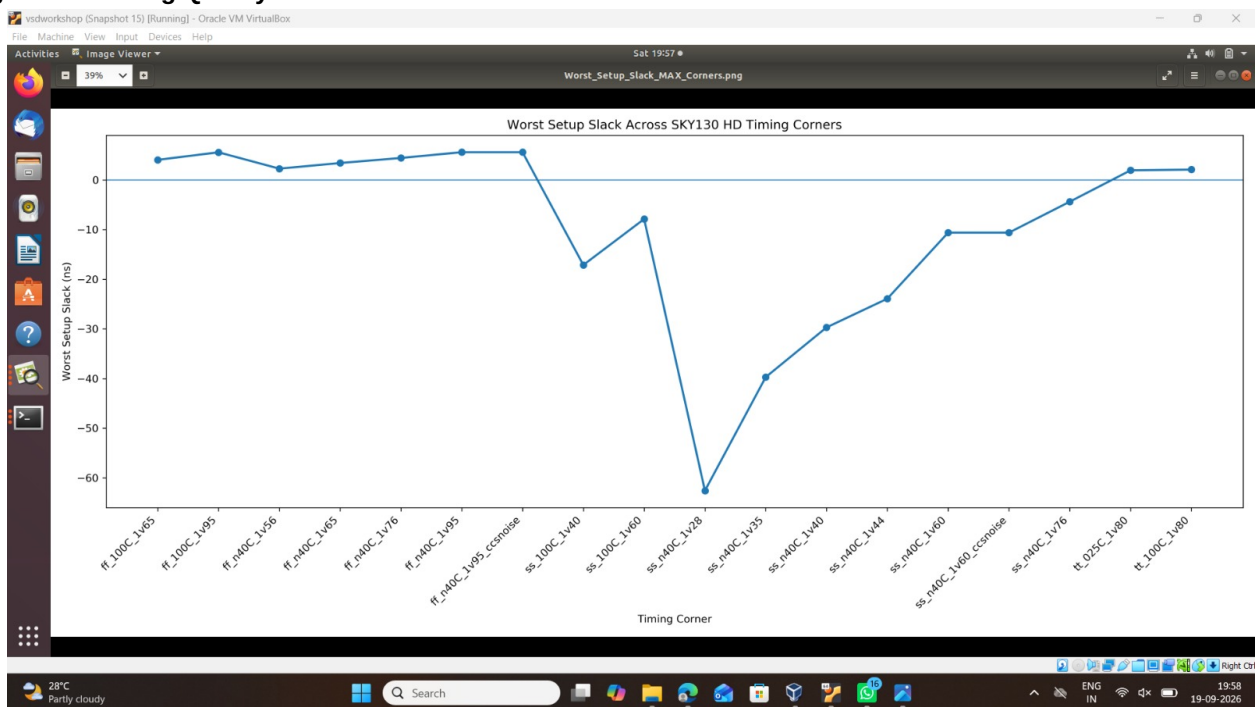
Extracted WNS/TNS values showing variation between fast and slow corners.

Figure 13. Completion of WNS/TNS Extraction



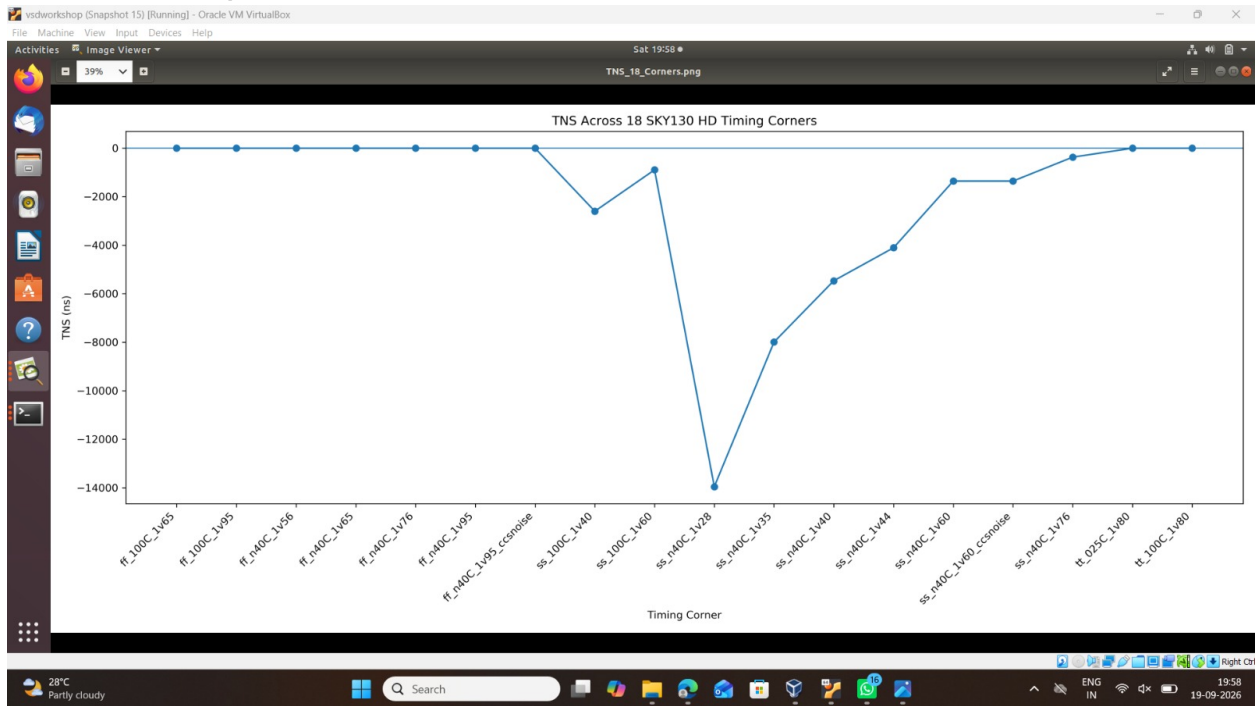
Completion of timing-metric extraction for the full 18-corner set.

Figure 14. Timing Quality Plot Generation



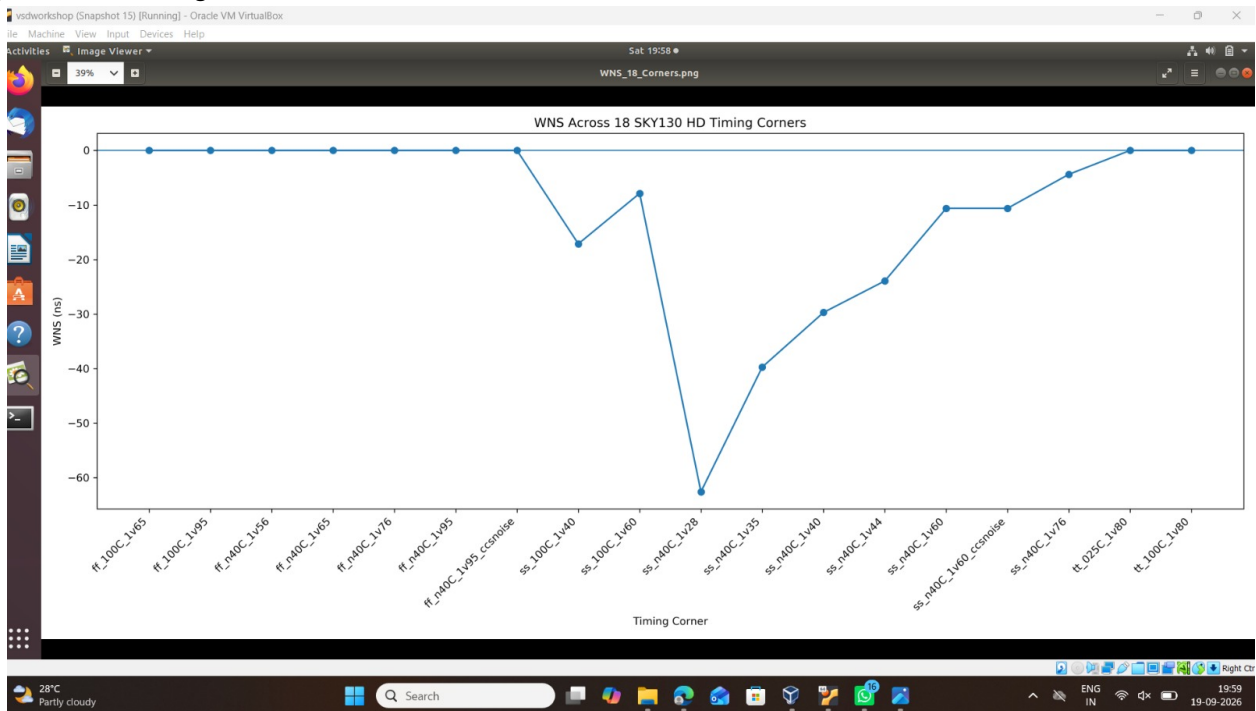
Generated WNS, TNS and hold-slack graphs for visual comparison.

Figure 15. Worst Setup Slack Across 18 SKY130 HD Corners



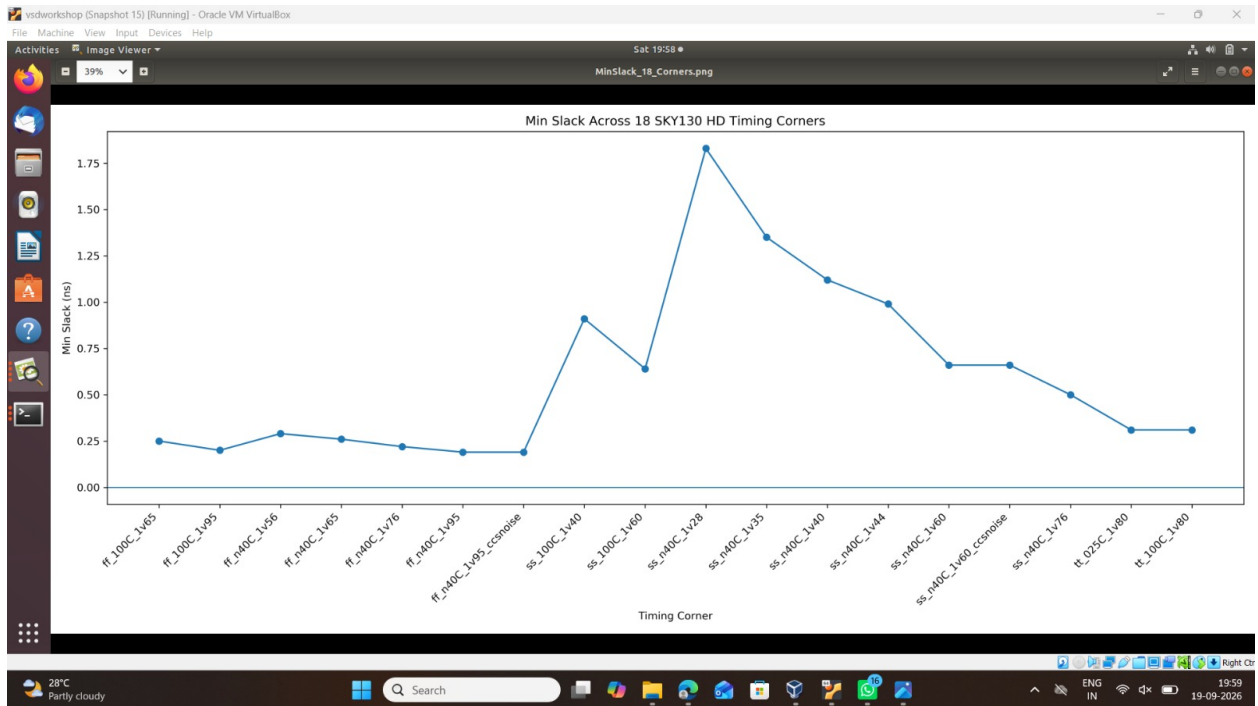
Worst setup slack plot; the supplied analysis reaches about -62.6 ns at SS n40C 1.28 V.

Figure 16. Total Negative Slack Across 18 Corners



TNS plot; the supplied analysis reaches about -13,555.8 ns at SS n40C 1.28 V.

Figure 17. WNS and Minimum Hold Slack Across 18 Corners



WNS and minimum-hold-slack views; displayed hold slack values remain positive.

6. Observations

The minimum-delay checks are positive for all listed corners, indicating that the reported hold paths have non-negative slack.

The maximum-delay results show positive slack for the FF and TT corners in the generated reports. The SS corners contain negative maximum-path slack values, indicating setup timing violations under those timing conditions.

The most negative maximum-path slack in the reported results occurs at the SS -40°C, 1.28 V corner.

7. Conclusion

Static Timing Analysis of the VSDBabySoC synthesized design was performed using OpenSTA and SKY130 HD timing libraries. Multiple process-voltage-temperature corners were analyzed using minimum and maximum timing paths.

The analysis provides visibility into hold and setup timing behavior across different operating conditions. The detailed STA reports and graphs can be used to identify timing-critical corners and paths for further optimization.